

Heidi Liu

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Education

University of Arizona, Tucson, AZ

Doctor of Philosophy in Physics; GPA: 3.5

August 2024 - Present

University of Washington, Seattle, WA

Bachelor of Science in Astronomy and Bachelor of Science in Physics; GPA: 3.61

March 2023

Graduated with Honor in Astronomy, Dean's List, Ralph Leonard Achievement Scholarship Recipient

Green River College, Auburn, WA

Associate in Science and Associate in Art; GPA: 3.91

August 2020

Graduate with the Highest Honor, Phi Theta Kappa Honor Society, Dean's List

Experience

University of Arizona, Astronomy and Physics Department

Revised UV Luminosity Function for $R_h = ct$ Universe, August 2025 – Present

Advisor: Prof. Fulvio Melia

- Built an end-to-end cosmology Python pipeline: linear transfer function $\rightarrow$ matter power spectrum $\rightarrow$ halo mass function $\rightarrow$ UV luminosity function. And tested Λ CDM against alternative expansion histories
- Reproduced literature HMFs and recalibrated published datasets to a common cosmology.
- Forecasted Converted halo growth $\rightarrow$ star-formation and dust-attenuated UV output using published, pre-JWST “consensus” prescriptions; quantified sensitivity to star-formation efficiency and intrinsic scatter.
- Produced fully reproducible notebooks (readme, fixed random seeds, figure scripts); organized results for advisor review and future manuscript drafting.

GSI Helmholtz Centre for Heavy Ion Research, Theory Department

DAAD (German Academic Exchange Service) RISE Professional, June 2023 – January 2024

Advisor: Dr. Thomas Neff

- Employed Fortran to calculate the nuclear shell model as input for astrophysical processes.
- Developed Python code to generate data of the Shell Model and create databases for the output files.
- Utilized Python for in-depth analysis of the shell model calculation data to study properties of heavy elements' decay in astrophysical processes.

University of Washington, Astronomy and Physics Department

CMB-S4 (Cosmic Microwave Background – Stage IV), March 2022 – June 2023

Advisor: Prof. Marilena Loverde

- Used the public Boltzmann code CLASS to compute the Cosmic Microwave Background power spectra and study how sources of new physics modify these power spectra.
- Developed methods in Python to simulate the primordial power spectra with tensor-to-scalar ratio.
- Forecasted constraints on the primordial gravitational wave using Fisher Matrix and MCMC chain.
- Attended weekly Snowmass paper meetings to review different topics in Cosmology in 2022.

University of Washington, Astronomy and Physics Department

DAMIC-M (Dark Matter In CCDs at Modane), January 2021 – January 2022

Advisor: Prof. Alvaro Chavarria

- Developed methods to reduce electrical noise for the Charge-Coupled Devices.
- Collaborated with Linux systems and Arduino to run tests for different components of the CCD's systems.
- Collaborated with Python and ROOT to analyze the CCD image and to improve noise distribution.

- Evaluated the charge collection efficiency of the CCD with image analysis.
- Compared the performance of a CCD packaged with a pitch adapter to the baseline performance of a CCD in the traditional package.

University of Washington, Manastash Ridge Observatory

AUEG (Astronomy Undergraduate Engineering Group), October 2020 – June 2023

Advisor: Prof. Oliver Fraser

- Updated the software system of the telescope and did maintenance work for the Observatory.
- Observed at the observatory for the advanced-level observation class projects (Standard & Variable Star).
- Attended the biweekly AUEG club meetings to update progress.
- Ran tests with fiber optics characterization.
- Developed plans to detect the Very low frequency during the solar eclipse using the High-Altitude Balloon.

Technical Skills

Computer Languages & Data: Python, Linux & Unix operating systems, SQLite, ROOT.

Others: SAOImageDS9, LaTeX, Jupyter Notebook, Mathematica, and Microsoft Office.

Languages: Chinese (Native), English (Fluent), German (Beginner).

Leadership / Service

University of Arizona Physics Department | Tucson, Arizona

Graduate Teaching Assistant, August 2024 – Present

- Led weekly PHYS 141 Mechanics / PHYS 182 Electromagnetism laboratory sections, guiding students through experimental design, data collection, and analysis.
- Graded lab worksheets and reports, providing feedback on data analysis, and scientific writing.
- Maintained course grade records, communicated lab policies and deadlines, and graded exams for over 300 students in the introductory physics course.

University of Washington Astronomy Department | Seattle, Washington

Undergraduate Astronomy Course Grader, September 2021 – March 2023

- Graded assignments and examinations for over 300 students in the introductory Astronomy class per quarter.
- Graded for the upper-level ASTR 301-Astronomy for Scientists and Engineers class.
- Proofread quizzes and homework problems and maintained gradebooks.

Particle Physics and Cosmology Snowmass Meeting | Seattle, Washington

Snowmass Outreach Coordinator, July 2022

- Taught the general public about the cloud chamber and particle physics.
- Taught grade-level students about gravitational waves and dark matter.

Presentations

Symposium, “*The Cosmic Timeline implied by the High- z Galaxy UVLF Function.*”

Steward Observatory Astronomy Symposium, Tucson, Arizona, March 2026.

Colloquium, “*Nuclear Shell Model Calculations for Astrophysical Processes.*”

GSI Helmholtz Centre for Heavy Ion Research, Darmstadt, Germany, September 2023.

Talk “*Forecasting Constraints on Primordial Gravitational Waves.*”

CMB-S4 Spring 2023 Collaboration Meeting, April 2023.

Poster, “*Forecasting Constraints on Primordial Gravitational Waves.*”

241st American Astronomical Society Meeting, Seattle, WA, January 2023.

Conference for Undergraduate Women in Physics, WA, January 2023.

Public Outreach, “*Observe the Universe across 30 orders of Magnitudes.*”

Theodor Jacobsen Observatory, Seattle, WA, August 2022.